ML Assignment – 02

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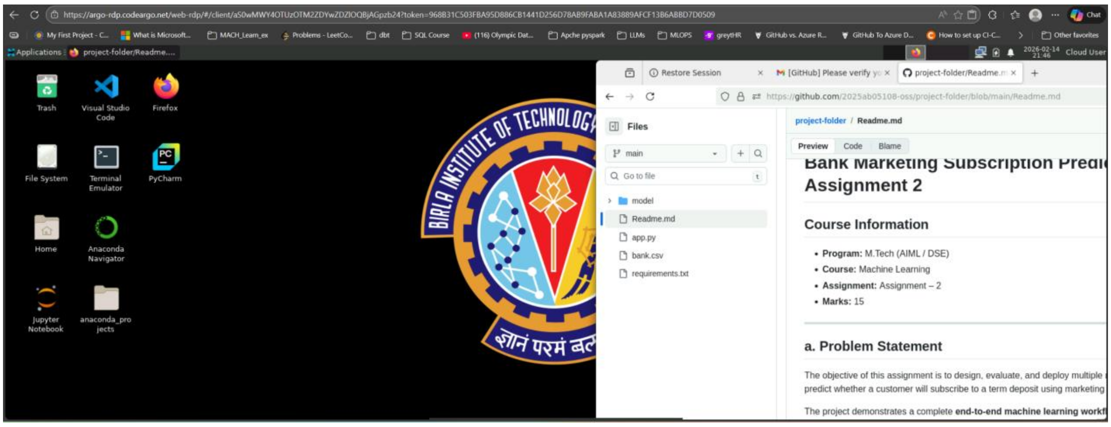
Subject - Machine Learning (S1-25_AIMLCZG565)

Artifacts:

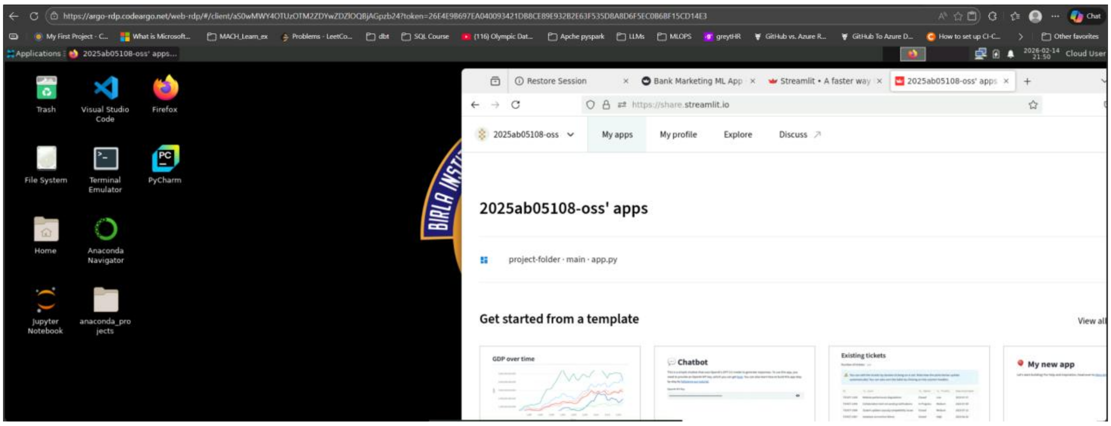
1. GitHub URL - **https://github.com/2025ab05108-oss/project-folder**
2. Streamlit.io URL - https://share.streamlit.io/
3. StreamLit.io Application - https://project-folder-iucedghan984k7bqdrufkx.streamlit.app/

Screenshots (Proof)

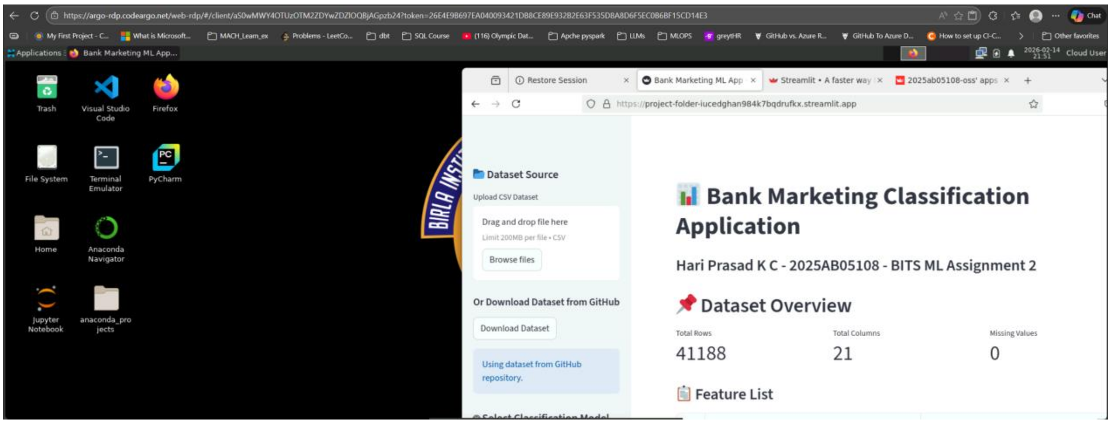
Screenshot 1 – GitHub



Screenshot 2 – Steamlit Dashboard



Screenshot 3 – Steamlit Application



<u>Readme.md File</u>

Bank Marketing Subscription Prediction – ML Assignment 2

Course Information

- **Program:** M.Tech (AIML / DSE)
- **Course:** Machine Learning
- **Assignment:** Assignment – 2
- **Marks:** 15

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a. Problem Statement

The objective of this assignment is to design, evaluate, and deploy multiple machine learning classification models to predict whether a customer will subscribe to a term deposit using marketing campaign data.

The project demonstrates a complete **end-to-end machine learning workflow**:

- Dataset preprocessing and encoding
- Implementation of multiple classification models
- Evaluation using standard performance metrics
- Deployment of an interactive Streamlit web application

This is a **binary classification problem**, where:

- `1` → Customer subscribed
- `0` → Customer did not subscribe

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b. Dataset Description [1 Mark]

- **Dataset Name:** Bank Marketing Dataset
- **File Used:** `bank.csv`
- **Problem Type:** Binary Classification

Dataset Characteristics

- **Number of Instances:** 41,188
- **Number of Features:** 21
- **Target Variable:** `y`
  - `1` → Subscription
  - `0` → No Subscription

Data Preprocessing & Cleansing

As implemented in `app.py`, the following preprocessing steps were applied **before model training**:

- Label encoding for categorical variables
- Train-test split (80:20 ratio)
- Feature scaling using **StandardScaler** (for Logistic Regression and KNN)

These steps ensure realistic model evaluation and prevent data leakage.

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c. Models Used & Evaluation Metrics [6 Marks]

All models were trained and evaluated on the **same processed dataset**.

Implemented Models

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbors (KNN)

4. Naive Bayes (Gaussian)
5. Random Forest (Ensemble Model)
6. XGBoost (Ensemble Model)

Evaluation Metrics

Each model was evaluated using:

- Accuracy
- AUC Score
- Precision
- Recall
- F1 Score
- Matthews Correlation Coefficient (MCC)

Model Comparison Table

ML Model	Accuracy	AUC	Precision	Recall	F1 Score	MCC
Logistic Regression	0.91	0.91	0.67	0.42	0.51	0.51
Decision Tree	0.89	0.89	0.51	0.51	0.51	0.51
KNN	0.90	0.90	0.59	0.40	0.47	0.47
Naive Bayes	0.85	0.85	0.40	0.61	0.48	0.48
Random Forest	0.91	0.91	0.65	0.51	0.57	0.57
XGBoost	0.92	0.92	0.65	0.55	0.60	0.60

Observations on Model Performance [3 Marks]

ML Model	Observation
Logistic Regression	Provides a strong baseline accuracy (91%) but shows low recall (0.42) for the minority class.
Decision Tree	Balanced performance but slightly lower accuracy (89%) and may overfit compared to ensemble models.
KNN	Moderate performance; lower recall indicates difficulty detecting positive class instances.
Naive Bayes	Lowest overall accuracy (85%) but highest recall

ML Model	Observation
	(0.61), detecting more subscribers.
Random Forest	Strong ensemble model with improved F1-score and better variance control.
XGBoost	Best overall performance (92% accuracy, highest F1-score 0.60) with balanced precision and recall.

Streamlit Application Features

The Streamlit application includes:

- CSV dataset upload option
- GitHub dataset auto-download option
- Model selection dropdown
- Display of all required evaluation metrics
- Confusion matrix visualization
- Classification report display

Project Structure

```
project-folder/
| -- app.py
| -- requirements.txt
| -- README.md
| -- bank.csv
| -- model/
├── logistic_regression_model.py
├── decision_tree_model.py
├── knn_model.py
├── naive_bayes_model.py
├── random_forest_model.py
└── xgboost_model.py
```

How to Run the Application

Install Dependencies

pip install -r requirements.txt

Run Streamlit App

streamlit run app.py

Execution Environment

- The assignment was executed using Python 3.10
- Deployment performed on Streamlit Community Cloud

Academic Integrity Declaration

This assignment has been independently implemented in accordance with the Academic Integrity Guidelines. AI tools were used only for conceptual understanding and learning support.

Submission Details

Platform Used: Streamlit Community Cloud

Python Version Used: 3.10

Deployment Type: Free Tier

GitHub Repository: https://github.com/2025ab05108-oss/project-folder

Live Application URL: https://project-folder-iucedghan984k7bqdrufkx.streamlit.app

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